Land Care Dashboard Documentation

Introduction

These dashboards provide insight into the parcel upkeep, check requests, and budgeting associated with the URA's land care organizations. Data used in these dashboards are regularly pulled from Regrid and NetSuite, and they are expected to be updated at the start of every monthly period. Detailed below are the methods used to create, maintain, and use the dashboards.

Links

Asset Management Homepage:

Link

Dashboard on Power BI Service:

Link (Private)

Link (Public)

SharePoint File Location:

Link

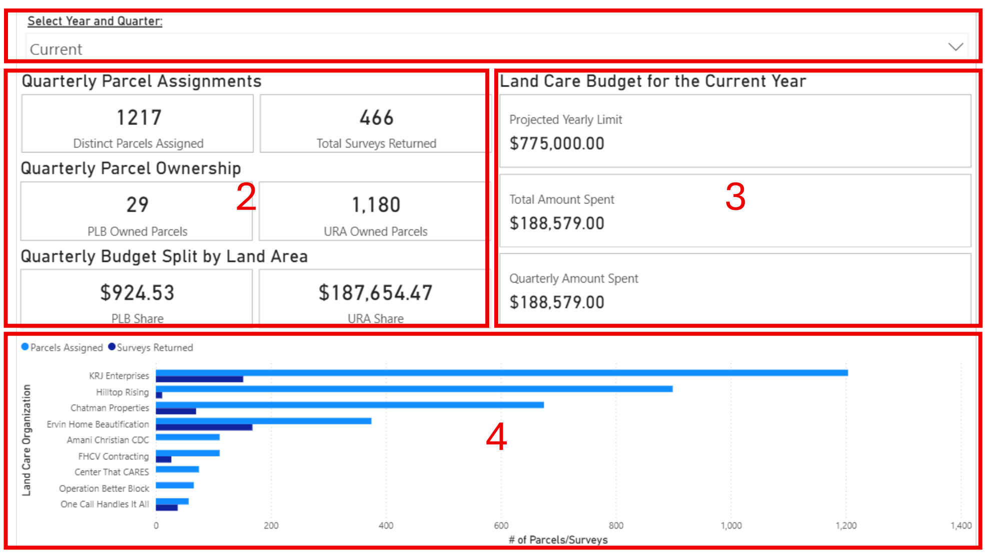
Dashboard Pages

The dashboard contains six pages, each providing in-depth statistics and information pertaining to land care and parcel distribution. The following section of the documentation breaks down each dashboard into its major components, explaining what kinds of information can be gleaned from them.

NOTE: THIS SECTION IS UNFINISHED

Landing Page

Viewers are automatically taken to the landing page upon opening the dashboard. This page contains general information such as quarterly land parcel statistics, and an overview of the current land care budget.



1. Filter

By selecting a specific year and quarter using this filter, users can change the time frame for which data is displayed on the page. The filter defaults to the current quarter upon loading the page.

2. Quarterly Statistics

This section displays high-level statistics on the land care parcels assigned during a given quarter. The first row displays the number of distinct parcels assigned overall, along with the total number of Regrid surveys that have been submitted for those parcels throughout the same period. The second row shows how many parcels are owned by the URA and the Pittsburgh Land Bank, separately. Finally, the third row uses the proportion of land area in the second row to calculate how much of the land care budget spent ___ is allocated to each ___.

3. Land Care Budget for the Current Year

This section provides a brief overview of the land care budget for the current year. It gives the overall yearly budget limit, how much has been spent thus far, and how much of that money was spent during the current quarter specifically.

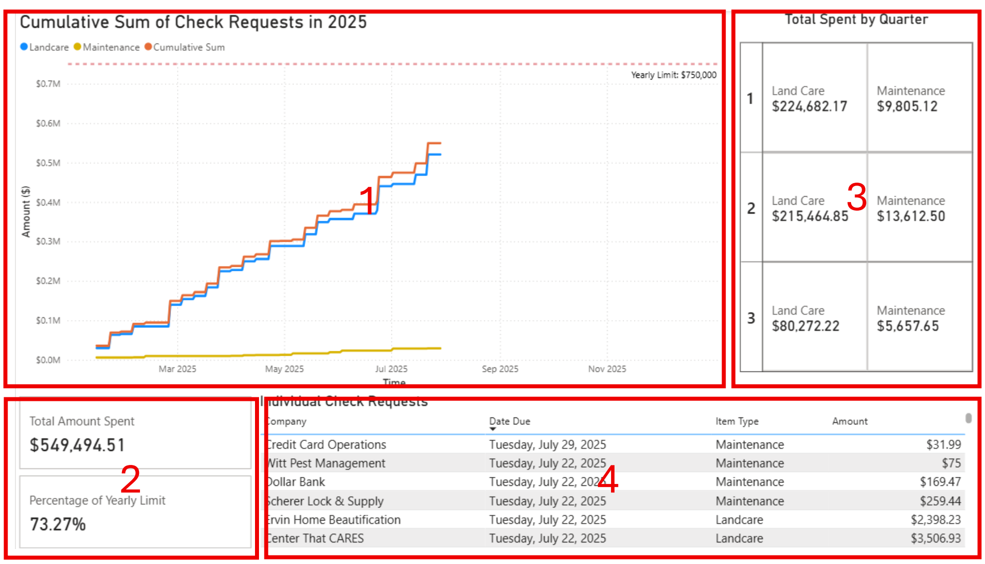
It should be noted that this section is not affected by the filter, and will always display the budget amounts for the current year and quarter.

4. Land Care Parcel/Survey Count

This section illustrates the values provided in component 2. Data is split among the eight main land care companies working with the URA, showing how many distinct parcels and returned surveys are associated with each in the given fiscal quarter.

2025 Budget Snapshot

The second dashboard delves more into the details of the land care budget throughout 2025, with a special focus on the individual check requests that make up the expenses. Values on this page are generally split between "land care" and "maintenance" types, providing more insight on the budget's allocation. While the values are dynamic, they will only continue as such up to the end of 2025.



1. Cumulative Sum of Check Requests in 2025

The first section is a line graph of the budget spent throughout the year, split into multiple lines for land care, maintenance, and a combination of both. As more payments are made, the line will continue to extend towards the static limit of $750,000 along the top of the graph. Hovering over any point along the lines will allow the user to see the values for that day, and clicking on that point will filter the measurements in the other components of the dashboard to help with precise calculations.

2. *Total Amount and Percentage*

This section provides some overview statistics on the land care budget. The top row shows the total amount spent on check requests as of the most recent date, while the bottom row calculates what percentage that value comprises of the yearly limit.

3. *Total Spent by Quarter*

This section splits check request payments by fiscal quarter, illustrating how much has been spent on both land care and maintenance during each segment. As payments are added in future quarters, the component will automatically update to include those quarters.

4. *Individual Check Requests*

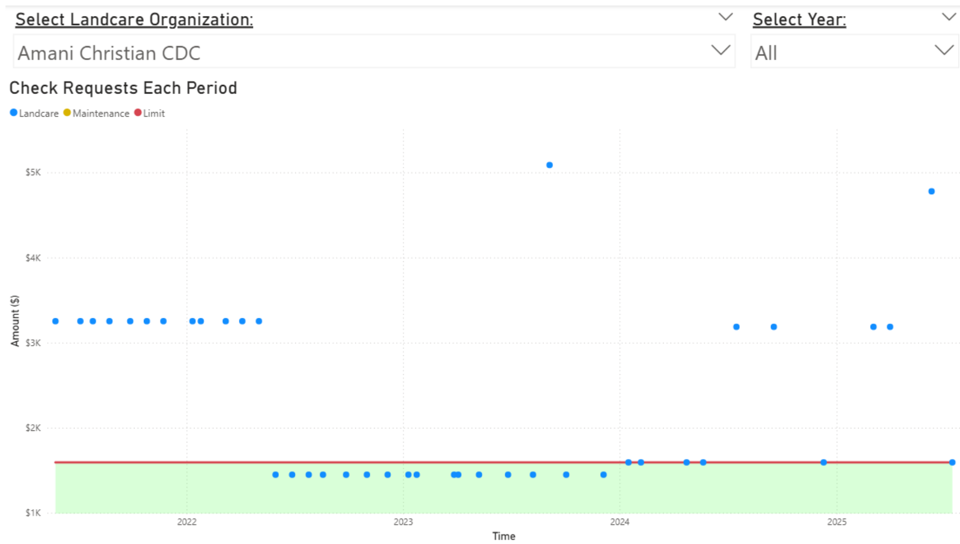
The final section is a comprehensive table of all check requests made throughout 2025. The table lists the company that received the payment along with the amount sent, the date of transaction, and whether the payment was for land care or maintenance. As mentioned above, clicking on a specific date from the line graph will filter this table to only include the check requests made on that day.

Check Request History

This dashboard provides a more in-depth look at the individual check requests made by each land care organization. Rather than a cumulative sum represented as a line graph, the data is represented as a scatterplot of individual check requests.

Each land care company has its own limit on the maximum amount of money that can be received per monthly check request. This limit changes between several periods, which is captured by the fluctuating line imposed on the scatterplot. Hovering over each point will show the invoice amount and how it compares to the limit for that date.

By using the filters across the top of the dashboard, the user can change which land care company and year(s) are being displayed. This naturally alters what data points are displayed, but it also shifts the line representing the invoice limit.

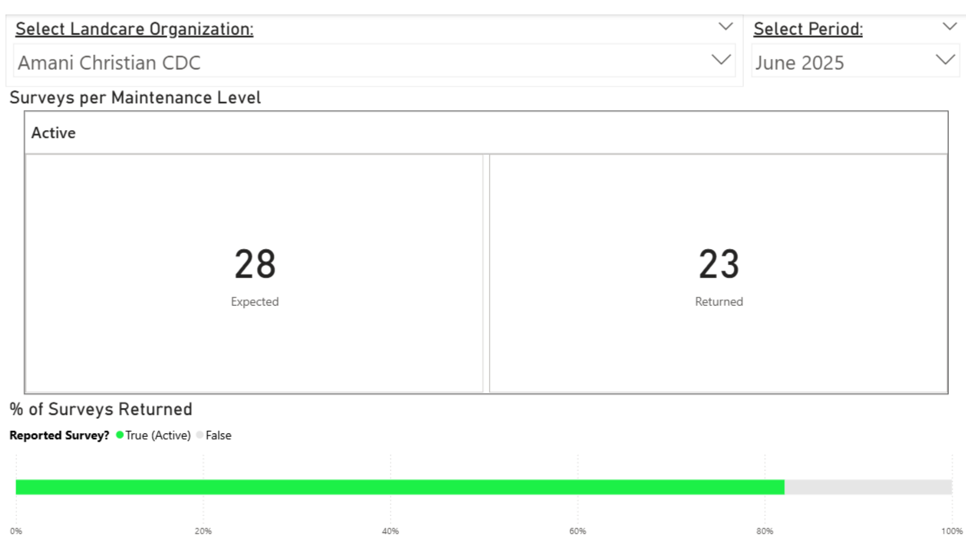


Survey Submission Rate

As land care organizations fulfill their responsibilities, they must also fill out surveys explaining what services have been provided to each parcel. This dashboard tracks the number of survey submissions associated with each land care organization per period – both the expected number, and what was actually returned.

Parcels can be categorized by the approach taken to maintain them. Some are categorized as “Active” and will be visited every period. Others are categorized as “Request Only” and will only be visited if the owner contacts the land care organization. The dashboard distinguishes between the two types of land when providing both flat numbers and percentages, shedding insight on how diligent a company is with reporting back.

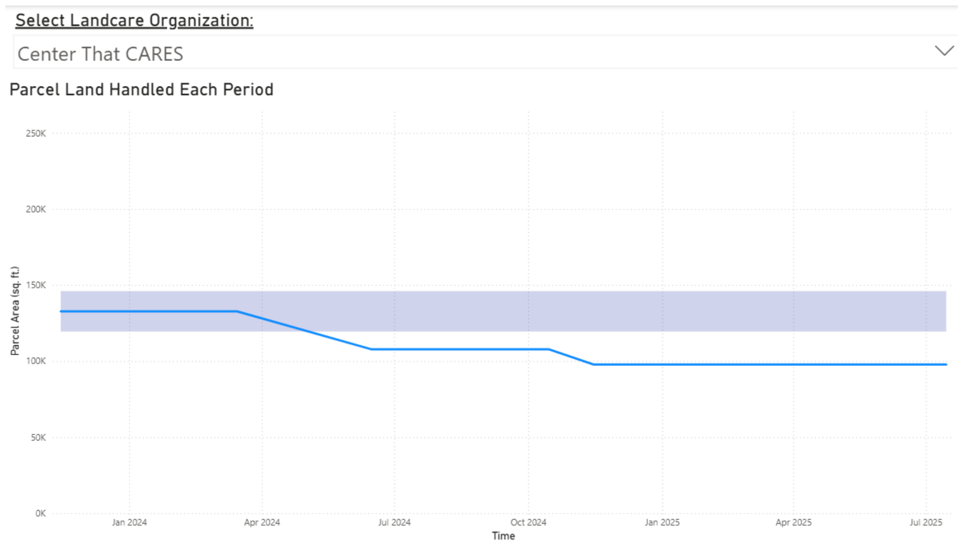
Additionally, this page also has filters for land care organization and upkeep period.



Parcel Land Area Distribution

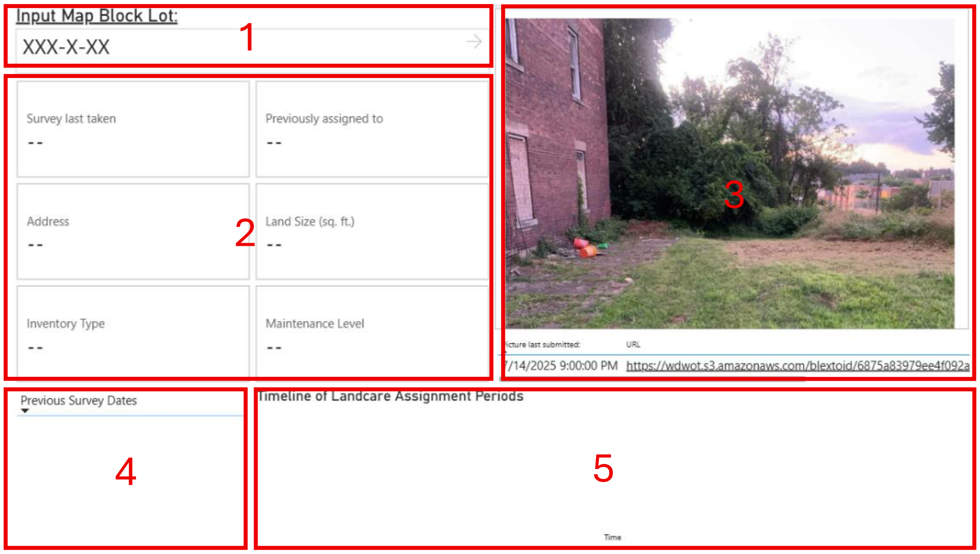
All land care organizations working with the URA have signed a contract, stating that the total land area they are responsible for will not fluctuate between +/- 10% of the area they started with. This dashboard is meant to affirm this promise by plotting measurements of area during each land care period. The line graph indicates such, while the highlighted area imposed on the graph represents the "safe" margin where assigned land lies within 10% of the original value.

Users can filter between each land care organization, displaying the parcel area measurements specific to that group while also dynamically changing the safe margin.



Parcel Land Care Details

This final dashboard provides a deeper look into the individual parcels, allowing users to search for specific parcels and returning any relevant information. The data gets updated according to the most recent Regrid survey submission for that respective parcel.



1. *Map Block Lot Search*

In the top-left corner of the dashboard lies a filter. Unlike the previous ones, this filter requires the user to type in a specific map-block-lot code. If the input matches a parcel that previously had its information submitted through a survey, then the other sections will update to show the most recent survey results. Otherwise, the results will be blank.

2. *Parcel Information*

This section provides the most up-to-date background information based on Regrid survey history. It displays the date of the most recent survey; the parcel's address, land size in square feet, inventory type, and maintenance level; and what company last looked at the property.

3. *Survey Photo*

Each land care survey requires at least one photograph of the cleaned property, in order to prove that the work has been completed. The most recent photograph is displayed here, along with the date it was taken and a URL to view the picture in full.

4. *Previous Survey Dates*

This section is a list of all previous dates when a survey was submitted for the filtered parcel. By clicking on one of these dates, the rest of the page will be filtered to display the information from that time, rather than the most recent date.

5. *Timeline of Landcare Assignment Periods*

Finally, this section presents a timeline of which land care organizations were in charge of looking after the parcel. The timespan of each company's tenure is presented along a singular line, while hovering over the line will reveal the individual dates when surveys were conducted. Clicking these dates will also filter the dashboard to that particular survey, just like the list of dates.

Important Processes

For the dashboard to update and display information properly, several steps must occur in a pipeline that creates and downloads data, inserts it into the URA's remote database, and queries it within the dashboard.

Creating Land Care Bundle Assignments

1. *Create bundle assignments for the new land care period*

On the 15th of each month at 3:00am, the virtual machine's Windows Task Scheduler is set to initiate a task called "Bundle Assignment Creation." This runs a Python script called "bundle_assignment_creation.py," which takes the current information in the database's EPP snapshot table and saves it as a new CSV file. This acts as the parcel assignments for the current land care period.

It should be noted that G Drive folder where the CSV file gets created is pre-defined within the Python script. **This means that when changing parameters for a new contracting period, the respective line of code will need to be adjusted.**

2. *Upsert bundle assignments to remote database*

Once the first task is finished, the virtual machine is set to immediately run a second task at 3:15am called "Upsert Bundle Assignments to Database." This runs the "BundlesDriveToSQL.py" Python script, which upserts the new parcel assignments from the previous step to the relevant table on the remote database.

Just like with the first task, the Python script also has a section of code where G Drive folders are defined to upsert the correct information. Unlike the previous script, however, this one contains multiple folder locations in a list. The code iterates through these folders to upsert potential changes in not just the current contracting period, but all previous periods. **When transitioning to a new contracting period, the address for the new bundle assignment folder must be added to this list.**

Additionally, the tables involved in bundle assignment creation are created using the script. **In a scenario where the database is wiped, this script must be run in order to reestablish all relevant tables.**

Downloading Regrid Survey Responses

1. *Download survey responses directly from Regrid*

Every 15th at 3:30am, the "Regrid Survey Download" task is run via the Windows Task Scheduler. This runs the "regrid_survey_download.py" script, which opens a new Microsoft Edge window on the virtual machine and emulates the process of navigating the website.

Something EXTREMELY IMPORTANT to note is that the URL for the Regrid surveys is not fixed. Each time a new survey is created for an upcoming period, a new export URL is also established. The existing code necessitates that this URL be changed manually. It may be possible to have this

Moreover, the script currently utilizes Oscar Medina's login information for Regrid. **This information should ideally be transferred to a more secure location,** such as a .env file.

2. *Upsert survey responses to remote database*

After giving a 3-day period to make room for any additional survey responses or corrections on Matthew Reitzell's part, Windows Task Scheduler will run the "Upsert Regrid Surveys to Database" task every 18th at 3.45am. This will run the "SurveysDriveToSQL.py" script, which upserts the new Regrid responses to the remote database.

The script once again lists out the G Drive folders that contain any Regrid surveys, and iterates through them when checking if there are any new additions that must be added. **This means that whenever the URA transitions to a new contracting period, the new folder location for downloaded surveys should be added to this list.**

Moreover, this script is responsible for creating the database tables related to survey responses. **In the scenario that the database is wiped or must be reset, this script should be run in order to reestablish the tables.**

Updating Land Care Budgeting and Contracts

All information on budgeting and contracts is taken directly from the "Land Care Annual Budgeting and Contracting.xlsx" file in the G Drive's public LandCare folder. The Python script "ContractsDriveToSQL.py" is responsible for upserting this information onto the database. This aspect of the pipeline is currently not automated. **As such, the script should be run manually whenever the Excel file is changed, or when the table must be reestablished. Additionally, the file's address should be updated within the code if changed within the drive.**

Creating NetSuite Reports

The dashboard also receives data directly from NetSuite, in addition to querying from the remote database. The reports used to obtain this data can be found by logging into the GIS NetSuite account, and navigating to **Reports > Saved Reports > All Saved Reports**. Additional reports may be created by saving copies as new files.

To connect a report with a dashboard, simply click the "Export – Excel Web Query" option in the bottom-right corner of the main view. This downloads a web link, which must be edited by inserting the GIS account's email and password. Finally, you just need to add a "Web Query" data source within the dashboard file and paste the link.

1. *All URA LandCare Check Requests*

This report queries various check requests made regarding land care efforts. It relies on fields within the Transactions item, limiting results to paid-in-full check requests created by the URA for land care and maintenance purposes, taking place on or after the year 2020.

2. *All URA LandCare Funding Requests*

This report queries any funding requests specifically for the current contracting period. It searches for instances where the Related Check Requests item is labeled as being a part of LandCare, and limits purchases to those taking place on or after the year 2024.

Refreshing Dashboard Data

Once all of this data is added to the dashboard, it gets refreshed daily within the semantic model on the Power BI online service – found here. In order to edit these parameters, you must be logged into the GIS Microsoft account.

Under "Gateway and cloud connections," there are parameters to ensure that the connection to the data sources remains secure. The GIS Data Gateway has already been configured to include credentials for the Postgres server and NetSuite. Clicking on the settings icon will allow you to change the details of important connections, as illustrated below.

Name ↑	Connection type	Users	Status	Gateway cluster name	Last credentials used
10.0.101.57;gisdb	PostgreSQL	GIS			
All Bundle Assignments	Folder	GIS		GIS Data Gateway	
All LandCare Check Requests	Web	GIS		GIS Data Gateway	4/29/26, 11:52 AM
https://1133549.app.netsuite.com/a...	Web	GIS			
Land Care Annual Budgeting and Co...	File	GIS		GIS Data Gateway	4/19/26, 8:17 PM
Regrid Survey Submissions	Folder	GIS		GIS Data Gateway	
URA VM	PostgreSQL	GIS		GIS Data Gateway	4/29/26, 11:52 AM

- All LandCare Check Requests – uses GIS NetSuite login credentials
- Land Care Annual Budgeting and Contracts – uses GIS Windows login credentials
- URA VM – uses Postgres login credentials